Linear Preferential Attachment in Directed Networks: Simulation, Maximum-Likelihood Estimation and Goodness-of-Fit Diagnostics

Numerical Experiments Report

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Abstract

This report presents the computational side of a project on the directed linear preferential attachment model of Bollobás, Borgs, Chayes and Riordan [1], with parameter estimation by maximum likelihood as developed by Wan, Wang, Davis and Resnick [2]. It describes a linear-time simulator of the model, a Monte Carlo check of the theoretical degree asymptotics, a study of the finite-sample behaviour of the likelihood estimators, and a fit-and-forecast exercise on four real temporal networks that compares the model with a trivial baseline. Figures and tables are read from the `out/` directory, where the accompanying code writes them.

1 Introduction

In a preferential attachment (PA) network, nodes that already hold many links are more likely to receive new ones. This simple rule produces heavy-tailed degree distributions of the kind seen in social, citation and communication data. For a statistician the directed linear PA model is convenient: when the growth history is observed its likelihood is explicit, and its limiting degree distribution is known exactly, so both estimation and sharp goodness-of-fit checks are possible.

Section 2 states the model and the theorem that the experiments probe; Section 3 derives the estimators; Section 4 gives the simulator (Algorithm 1); Section 5 collects the diagnostics; Sections 6 and 7 report the experiments; Section 8 summarises.

2 The model and the results under test

Model. In [1] a directed multigraph grows one edge at a time, so $G(t)$ has t edges and the process starts from a fixed graph with $t_0 \geq 1$ edges. Let $N(t)$ be the number of vertices and $D_v^{\text{in}}(t)$, $D_v^{\text{out}}(t)$ the degrees of v. For $\alpha, \beta, \gamma \geq 0$ with $\alpha + \beta + \gamma = 1$ and shifts $\delta_{\text{in}}, \delta_{\text{out}} > 0$, each step selects a scheme at random and adds a directed edge $v \to w$:

A (prob. α): v is a new vertex, and w is an existing vertex drawn with probability $\dfrac{D_w^{\text{in}}(t) + \delta_{\text{in}}}{t + \delta_{\text{in}} N(t)}$.

B (prob. β): v, w are existing vertices drawn independently, with probabilities $\dfrac{D_v^{\text{out}}(t) + \delta_{\text{out}}}{t + \delta_{\text{out}} N(t)}$ and $\dfrac{D_w^{\text{in}}(t) + \delta_{\text{in}}}{t + \delta_{\text{in}} N(t)}$.

C (prob. γ): w is a new vertex, and v is an existing vertex drawn with probability $\dfrac{D_v^{\text{out}}(t) + \delta_{\text{out}}}{t + \delta_{\text{out}} N(t)}$.

Loops and multiple edges are allowed. The extended model of [2] has further schemes creating two new vertices at once; they are omitted here, and Section 7 measures how often such edges occur in real data. Note that $N(t) = (\alpha+\gamma)t + O(t^{1/2}\log t)$ almost surely.

Degree asymptotics. Let $X_i(t)$, $Y_j(t)$ count vertices with in-degree i and out-degree j, and set

$$c_1 = \frac{\alpha+\beta}{1+\delta_{\text{in}}(\alpha+\gamma)}, \qquad c_2 = \frac{\beta+\gamma}{1+\delta_{\text{out}}(\alpha+\gamma)}. \tag{1}$$

Theorem 1 ([1])**.** *There are constants p_i, q_j with $X_i(t) = p_i t + o(t)$ and $Y_j(t) = q_j t + o(t)$ almost surely. If $\alpha\delta_{\text{in}} + \gamma > 0$ and $\gamma < 1$, then $p_i \sim c_{\text{in}}\, i^{-\tau_{\text{in}}}$ with $\tau_{\text{in}} = 1 + 1/c_1$; symmetrically $q_j \sim c_{\text{out}}\, j^{-\tau_{\text{out}}}$ with $\tau_{\text{out}} = 1 + 1/c_2$ (under $\gamma\delta_{\text{out}} + \alpha > 0$, $\alpha < 1$).*

The constants satisfy the explicit recursion

$$p_0 = \frac{\alpha}{1+c_1\delta_{\text{in}}}, \qquad p_1 = \frac{\delta_{\text{in}}p_0 + \gamma/c_1}{1+\delta_{\text{in}}+1/c_1}, \qquad p_i = \frac{i-1+\delta_{\text{in}}}{i+\delta_{\text{in}}+1/c_1}\,p_{i-1} \quad (i \ge 2), \tag{2}$$

and q_j is obtained by exchanging $\alpha \leftrightarrow \gamma$, $\delta_{\text{in}} \leftrightarrow \delta_{\text{out}}$. The tail of the cdf then decays as $k^{-(\tau-1)}$. The simulations in Section 6 test the exact constants at small degrees, the tail exponent, and the size of the error term.

3 Maximum-likelihood estimation

Assume the initial graph and the time-ordered edges $e_t = (v_t, w_t)$, $t_0 < t \le T$, are observed. The scheme J_t is then identifiable (new source: A; new target: C; neither: B). We estimate $\theta = (\alpha, \beta, \delta_{\text{in}}, \delta_{\text{out}})$, with $\gamma = 1-\alpha-\beta$.

Likelihood. The log-likelihood is additively separable, $\ell(\theta) = \ell_1(\alpha,\beta) + \ell_2(\delta_{\text{in}}) + \ell_3(\delta_{\text{out}})$, where $\ell_1 = \sum_t \left[\mathbb{1}_{\{J_t=A\}}\log\alpha + \mathbb{1}_{\{J_t=B\}}\log\beta + \mathbb{1}_{\{J_t=C\}}\log\gamma\right]$ and

$$\ell_2(\lambda) = \sum_{t=t_0+1}^{T} \Big[\log\big(D^{\text{in}}_{w_t}(t-1)+\lambda\big) - \log\big(t-1+\lambda N(t-1)\big)\Big]\mathbb{1}_{\{J_t\in\{A,B\}\}},$$

with ℓ_3 its out-degree analogue (over $J_t \in \{B, C\}$). The three parts are maximised independently.

Estimators. The maximisers of ℓ_1 are the scheme frequencies $\hat\alpha = \frac{1}{T-t_0}\sum_t \mathbb{1}_{\{J_t=A\}}$ and $\hat\beta = \frac{1}{T-t_0}\sum_t \mathbb{1}_{\{J_t=B\}}$. For the shifts, in-degrees increase in unit steps, so the sum of $\log(D^{\text{in}}_{w_t}(t-1)+\lambda)$ over time can be rewritten as a sum over degree levels. This gives the profile

$$\begin{aligned}\Psi_T(\lambda) = \frac{1}{T-t_0}\Bigg[&\sum_{i\ge 0}\big(X_{>i}(T) - X_{>i}(t_0)\big)\log(i+\lambda) - \Big(\sum_t \mathbb{1}_{\{J_t=C\}}\Big)\log\lambda \\ &- \sum_t \mathbb{1}_{\{J_t\in\{A,B\}\}}\log\big(t-1+\lambda N(t-1)\big)\Bigg],\end{aligned} \tag{3}$$

where $X_{>i}(t) = \sum_{k>i} X_k(t)$, and $\hat\delta_{\text{in}} = \arg\max_{\lambda\in[\varepsilon,K]}\Psi_T(\lambda)$. The estimator $\hat\delta_{\text{out}}$ is the maximiser of the analogous function Φ_T built from $Y_{>j}$ and $\hat\alpha$, and both maximisers are found numerically (Brent's method). Only the scheme sequence and the degree tails at times t_0 and T are needed, so a single pass over the edge list and memory linear in the number of vertices suffice.

Asymptotics [2]. For $\alpha, \beta, \gamma \in (0,1)$ and $\delta_{\text{in}}, \delta_{\text{out}} \in (\varepsilon, K)$,

$$\sqrt{T}\,(\hat{\theta}_T - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}), \qquad I(\theta)^{-1} = \operatorname{diag}\Big(\begin{pmatrix} \alpha(1-\alpha) & -\alpha\beta \\ -\alpha\beta & \beta(1-\beta) \end{pmatrix},\ I_{\text{in}}^{-1},\ I_{\text{out}}^{-1} \Big), \tag{4}$$

where the scalars $I_{\text{in}}, I_{\text{out}}$ depend on θ through p_i and q_j (explicit formulas in [2]). Hence the MLE is asymptotically efficient and the blocks $(\hat{\alpha}, \hat{\beta})$, $\hat{\delta}_{\text{in}}$, $\hat{\delta}_{\text{out}}$ are asymptotically independent; the experiments check both properties and the resulting confidence intervals.

4 Simulation algorithm

Algorithm 1 is the sampler of [2]. Edges are stored in two arrays indexed by creation time. The routine SampleVertex returns v with probability $(D_v + \delta)/(t + N\delta)$ in $O(1)$ time: with probability $t/(t + N\delta)$ it returns the relevant endpoint of a uniformly chosen edge (hitting v with probability D_v/t), otherwise a uniformly chosen vertex, so

$$\mathbb{P}(v) = \frac{t}{t + N\delta} \cdot \frac{D_v}{t} + \frac{N\delta}{t + N\delta} \cdot \frac{1}{N} = \frac{D_v + \delta}{t + N\delta}.$$

A network with n edges is thus generated in $O(n)$ time. The implementation is in Python with Numba and uses explicit seeds for reproducibility. Synthetic runs start from one vertex with a self-loop ($t_0 = N(t_0) = 1$), whose influence is negligible at 10^5–10^6 edges.

5 Diagnostic tools

Hill estimator. For ordered positive degrees $D_{(1)} \geq \cdots \geq D_{(M)}$ and a tail $\mathbb{P}(D \geq x) = L(x)x^{-\xi}$ (L slowly varying), the Hill estimator [3] is $\hat{\xi}_k^{-1} = \frac{1}{k}\sum_{i=1}^{k} \ln\big(D_{(i)}/D_{(k+1)}\big)$. Because the cdf tail index is $\tau - 1$, we report $\hat{\tau} = 1 + \hat{\xi}_k$, with $k = \lceil M^{1/2} \rceil$ (at least 50); a fixed-percentage rule would not be consistent. For model checking, $\hat{\tau}$ is compared with the indirect value $1 + 1/\hat{c}_1$ (or $1 + 1/\hat{c}_2$) from the MLE and (1).

Other tools. The Kolmogorov–Smirnov distance $\sup_x |F_1(x) - F_2(x)|$ compares degree distributions of two graphs. Normality is tested with Shapiro–Wilk. A sample correlation over R replications has null standard deviation $\approx 1/\sqrt{R}$, so $|r| > 1.96/\sqrt{R}$ is flagged as significant. Coverage of nominal 95% intervals is judged against the binomial band $\pm 1.96\sqrt{p(1-p)/R}$.

6 Experiments on synthetic networks

6.1 Limiting degree frequencies

Setup. We take $(\alpha, \beta, \delta_{\text{in}}, \delta_{\text{out}}) = (0.3, 0.5, 2, 1)$, the configuration also used in the simulations of [2], and generate $R = 200$ networks with $T = 10^5$ edges. Averaged frequencies $X_i(T)/T$ and $Y_j(T)/T$ for $i, j \leq 20$ are compared with the exact constants p_i, q_j from (2), not with their power-law approximation, so the check covers the full first statement of Theorem 1.

Finding. In Figure 1 every point falls on the theoretical curve within its error bar, for both in- and out-degrees and across all degrees examined, including the boundary values $i = 0, 1$ generated directly by schemes A and C.

Algorithm 1: Generation of the directed linear preferential attachment network

Input: parameters $\alpha, \beta, \delta_{\text{in}}, \delta_{\text{out}}$; initial graph $G(t_0) = (V, E)$ with $t_0 \geq 1$ edges; target number of edges n.

Output: graph $G(n) = (V, E)$.

$t \leftarrow t_0$;
while $t < n$ **do**
  $N \leftarrow |V|$; draw $U \sim \text{Unif}(0, 1)$;
  if $U < \alpha$ **then** // scheme A
    $v^{\text{out}} \leftarrow N + 1$; // new vertex is the source
    $v^{\text{in}} \leftarrow$ `SampleVertex`$(E, \text{in}, \delta_{\text{in}}, t, N)$;
    $V \leftarrow V \cup \{N + 1\}$;
  else if $U < \alpha + \beta$ **then** // scheme B
    $v^{\text{out}} \leftarrow$ `SampleVertex`$(E, \text{out}, \delta_{\text{out}}, t, N)$;
    $v^{\text{in}} \leftarrow$ `SampleVertex`$(E, \text{in}, \delta_{\text{in}}, t, N)$;
  else // scheme C
    $v^{\text{out}} \leftarrow$ `SampleVertex`$(E, \text{out}, \delta_{\text{out}}, t, N)$;
    $v^{\text{in}} \leftarrow N + 1$; // new vertex is the target
    $V \leftarrow V \cup \{N + 1\}$;
  $E \leftarrow E \cup \{(v^{\text{out}}, v^{\text{in}})\}$; // edge number $t + 1$
  $t \leftarrow t + 1$;
return $G(n) = (V, E)$;

Function `SampleVertex`(E, dir, δ, t, N)**:**
  Input: edge list E; direction $dir \in \{\text{in}, \text{out}\}$ (target or source endpoint); shift δ; current time t; number of vertices N.
  Output: vertex v with $\mathbb{P}(v) = (D_v^{dir} + \delta)/(t + N\delta)$.
  draw $W \sim \text{Unif}(0,\ t + N\delta)$;
  if $W \leq t$ **then**
    $v \leftarrow E^{dir}_{\lceil W \rceil}$; // dir-endpoint of edge $\lceil W \rceil$
  else
    $v \leftarrow \lceil (W - t)/\delta \rceil$; // uniform part: one of N vertices
  return v;

6.2 Tail exponents

Setup. For a handful of parameter sets, $R = 40$ networks with $T = 3 \cdot 10^5$ edges are simulated, and Hill estimates of τ_{in} (and τ_{out}) are compared with $1 + 1/c_1$ (resp. $1 + 1/c_2$).

Finding. Table 1 shows estimates close to theory for all parameter sets, with a consistent sign: they usually fall slightly short of the true exponent. This is a finite-size effect: the largest degree grows like t^{c_1}, so for small c_1 (large δ_{in}) the power-law window is short and the Hill estimator is biased downward. The first set, $(0.3, 0.5, 2, 1)$ with $c_1 = 0.4$, is the extreme case: against $\tau_{\text{in}} = 3.5$ the table gives 3.351 and 3.242, i.e. errors of about 4% and 7%. In contrast, $(0.41, 0.49, 0.30, 0.6)$ has a much larger c_1 and a wide window, and the estimate 2.281 is nearly the theoretical 2.253.

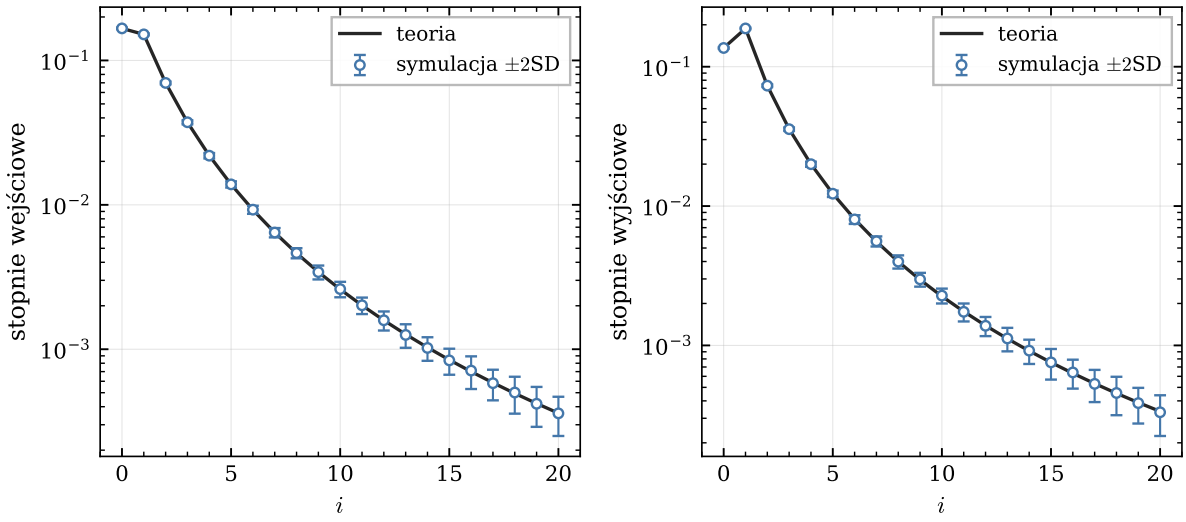


Figure 1: Average empirical frequencies $X_i(T)/T$ (left) and $Y_j(T)/T$ (right) over $R = 200$ networks, with bars of ± 2 between-network standard deviations, versus the theoretical p_i, q_j (line). Log scale on the vertical axis; $(\alpha, \beta, \delta_{\text{in}}, \delta_{\text{out}}) = (0.3, 0.5, 2, 1)$, $T = 10^5$.

Table 1: Comparing theoretical exponents τ_{in}, τ_{out} with Hill estimates with constraint $k = M^{1/2}$, where M is the number of vertices with positive degrees (mean $\pm$ standard deviation from $R = 40$ replications, $T = 3 \cdot 10^5$).

α	β	δ_{in}	δ_{out}	τ_{in}	$\hat{\tau}_{in}$ (Hill)	τ_{out}	$\hat{\tau}_{out}$ (Hill)
0.30	0.50	2.00	1.00	3.500	3.351 ± 0.100	3.143	3.079 ± 0.091
0.30	0.50	0.50	0.50	2.562	2.530 ± 0.077	2.786	2.747 ± 0.069
0.20	0.60	1.00	1.00	2.750	2.720 ± 0.075	2.750	2.706 ± 0.076
0.41	0.49	0.30	0.60	2.281	2.253 ± 0.061	3.214	3.136 ± 0.097
0.10	0.80	1.00	0.20	2.333	2.295 ± 0.063	2.156	2.128 ± 0.071

Row $(0.41, 0.49, 0.30, 0.60)$ matches the WWW graph calibration from paper [2].

6.3 Convergence rate of $X_i(t)/t$

Setup. The theorem gives $X_i(t) - p_i t = o(t)$, and its proof gives $O(t^{1/2} \log t)$. To see the actual size of the fluctuations, $R = 100$ networks with $T = 10^6$ edges are followed on a logarithmic time grid for $i \in \{0, 1, 2, 5, 10\}$ (boundary, intermediate and purely recursive cases of (2)). We plot $|X_i(t) - p_i t|$ divided by $\sqrt{t} \log t$ and, as a contrast, by $\sqrt{t}$; each curve begins when the expected count $p_i t$ passes a few dozen vertices.

Finding. With $\sqrt{t}$ scaling (right panel of Figure 2) the curves are flat over the whole window, whereas with $\sqrt{t} \log t$ they drift downward. The fluctuations are therefore of order $t^{1/2}$, possibly times a slowly varying factor. A polynomial correction, even as mild as $t^{0.1}$, would give a visibly positive slope in the right panel and is ruled out. All five degrees behave alike apart from a multiplicative constant: smaller populations fluctuate less in absolute terms.

Remark 1. The theorem is an upper bound and $\log t$ changes only about twofold over three decades, so the experiment cannot decide whether the logarithm is needed; it supports the order $t^{1/2}$ and excludes power corrections.

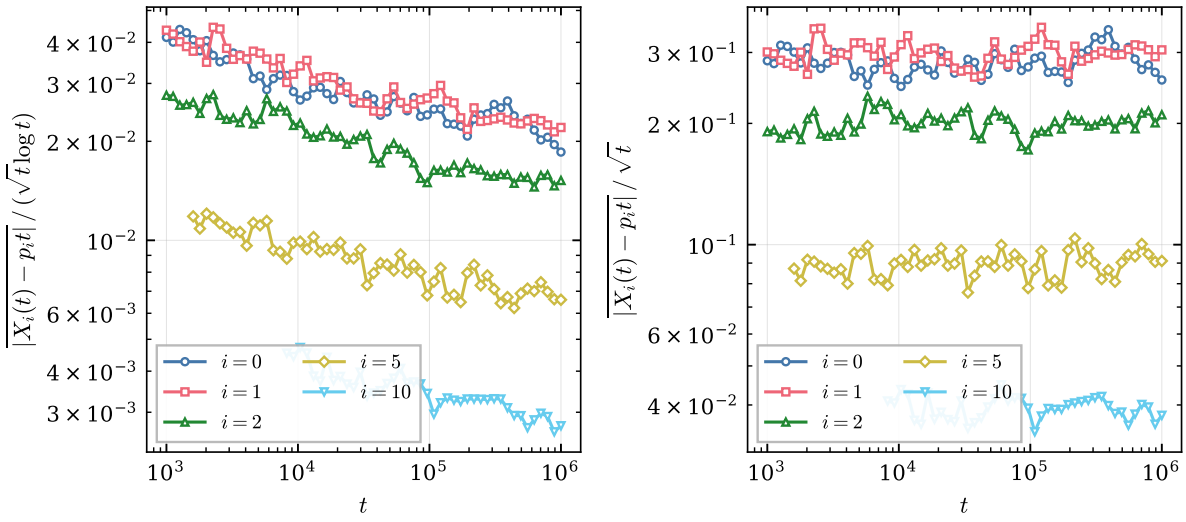


Figure 2: Mean absolute error $\overline{|X_i(t) - p_i t|}$ scaled by $\sqrt{t}\log t$ (left) and by $\sqrt{t}$ (right), $i \in \{0, 1, 2, 5, 10\}$; log–log axes, $R = 100$, $T = 10^6$, parameters as in Figure 1.

6.4 Properties of the MLE

Setup. The goal is to check that the implementation matches (4), that the block structure shows up in the sampling distribution, and from which size the intervals can be trusted. With $\theta = (0.3, 0.5, 2, 1)$ we simulate $R = 1000$ networks of $n = 10^5$ edges. In each, $\hat\alpha, \hat\beta$ are the scheme frequencies and $\hat\delta_{\text{in}}, \hat\delta_{\text{out}}$ maximise Ψ_T, Φ_T over $[\varepsilon, K]$ using Brent's method.

Bias and efficiency. The mean estimate is $\bar{\hat\theta} = (0.3000, 0.5000, 2.0008, 1.0011)$. Empirical standard deviations (0.00148, 0.00156, 0.02504, 0.01403) are within 4% of the asymptotic values (0.00145, 0.00158, 0.02412, 0.01418) given by $I(\theta)^{-1}/n$; the Cramér–Rao bound is thus essentially attained.

Block independence. Table 2 shows that every cross-block correlation lies in $[-0.038, 0.046]$,

Table 2: MLE estimates sample correlation matrix from $R = 1000$ replications ($n = 10^5$).

	$\hat\alpha$	$\hat\beta$	$\hat\delta_{in}$	$\hat\delta_{out}$
$\hat\alpha$	1.000	−0.651	−0.038	0.020
$\hat\beta$	−0.651	1.000	0.046	−0.031
$\hat\delta_{in}$	−0.038	0.046	1.000	0.033
$\hat\delta_{out}$	0.020	−0.031	0.033	1.000

below the $1.96/\sqrt{R} \approx 0.062$ threshold. Inside the block $(\hat\alpha, \hat\beta)$ the correlation is -0.651; theory, read off the upper-left block of (4), predicts $-\alpha\beta/\sqrt{\alpha(1-\alpha)\beta(1-\beta)} = -0.655$, and the gap of 0.004 is far smaller than the standard error of a correlation coefficient here (≈ 0.018). Both aspects of $I(\theta)$ are thereby confirmed: zero where it should vanish, and the predicted non-zero value where it should not.

Coverage. Nominal 95% intervals cover the truth with frequencies 0.945, 0.952, 0.935, 0.951 (standard error ≈ 0.007). The slight shortfall for $\hat\delta_{\text{in}}$ comes from estimating the Fisher information; with the exact σ the coverage is 0.941.

Onset of the normal regime. Repeating the experiment for $10^3 \le n \le 10^5$ with $R = 800$ replications per n, we record the Shapiro–Wilk p-value (for estimates standardised with the

theoretical σ, to isolate the distributional shape) and the coverage of intervals using a per-replication $\hat{\sigma}$.

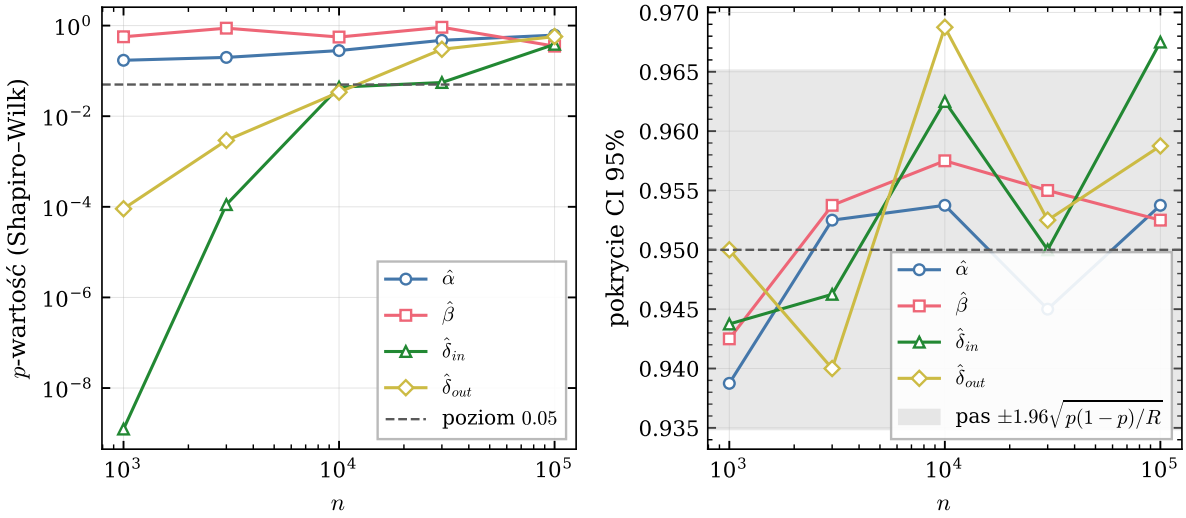


Figure 3: Approach to normality as a function of the number of edges n. Left: Shapiro–Wilk p-values by coordinate (log axis; dashed line at 0.05). Right: coverage of nominal 95% intervals; the grey band is the sampling-noise range $\pm 1.96\sqrt{p(1-p)/R}$. $R = 800$ per point.

Figure 3 separates the parameters into two groups. $\hat{\alpha}, \hat{\beta}$, being means of i.i.d. indicators, are normal from $n = 10^3$ onwards (p-values between 0.17 and 0.92). For $\hat{\delta}_{\text{in}}$ and $\hat{\delta}_{\text{out}}$, normality is rejected for $n \le 3 \cdot 10^3$ ($p \le 3 \cdot 10^{-3}$), is borderline at $n = 10^4$ ($p \approx 0.03$–0.04), and is no longer rejected from about $3 \cdot 10^4$. This is plausible, since the normality of $\hat{\delta}$ rests on a martingale CLT, which needs about ten times more data than an i.i.d. average. Coverage nevertheless stays correct over the whole grid. At $n = 10^3$ it equals 0.944 ($\hat{\delta}_{\text{in}}$) and 0.950 ($\hat{\delta}_{\text{out}}$), inside the band 0.95 ± 0.015, because two-sided symmetric intervals tolerate mild skewness.

7 Fitting the model to real networks

7.1 Data and estimation

Four directed temporal networks from SNAP [4] are used: `email-Eu-core` (e-mails inside a research institution), `CollegeMsg` (private messages on a student platform), `Reddit` (hyperlinks between subreddits, `soc-redditHyperlinks`) and `MathOverflow` (answers on a Q&A site). They range from almost closed populations to clearly expanding ones, and, as the results show, the speed of vertex growth is what determines how well the model fits.

Edges are ordered by timestamp, vertices are labelled in order of first appearance, and each edge is assigned to scheme A, B or C according to which endpoints are new. Edges whose two endpoints are both new (the additional scheme of the full model in [2]) are excluded from estimation; their share is small, from 0.37% (`email-Eu`) to 1.87% (`Reddit`), which supports the three-scheme specification. Self-loops matter only in `MathOverflow` (3.0% of edges, users answering their own questions) and are retained because the model admits loops.

Two regimes. Table 3 reports the MLE and the exponent check, with networks sorted by the combined share of schemes A and C (the rate of vertex growth). The communication networks are dominated by scheme B ($\hat{\beta} \approx 0.97$–1.00): their membership is essentially fixed and nearly every edge joins two existing members. In `Reddit` and `MathOverflow` new vertices are several times more frequent, which fits the growing-network premise of the model.

Table 3: Web characteristic (A/B/C scheme weights), MLE estimates, tail exponent consistency test (MLE w.r.t. Hill) and Kolmogorov-Smirnoff distance in predictive validation contrasted with base control ("nothing changes" prognosis). N is the ultimate vertices count.

web	N	A/B/C [%]	$\hat{\beta}$	$\hat{\tau}_{in}$ (MLE/Hill)	$\hat{\tau}_{out}$ (MLE/Hill)	KS_{in} (model/control)
`email-Eu`	985	0.1/99.7/0.2	0.997	2.01 / 4.21	2.00 / 2.95	0.235 / 0.081
`CollegeMsg`	1855	1.0/96.9/2.2	0.969	2.15 / 4.61	2.09 / 2.93	0.102 / 0.036
`Reddit`	33488	7.6/88.1/4.3	0.881	2.09 / 2.37	2.13 / 2.63	0.019 / 0.003
`MathOverflow`	20771	6.8/80.5/12.7	0.805	2.41 / 2.89	2.14 / 2.97	0.019 / 0.025

Exponent consistency. The tail exponent is estimated in two independent ways: indirectly as $1 + 1/\hat{c}_1$ from the MLE, and directly with the Hill estimator ($k = M^{1/2}$). If the model is right the two agree, so their difference is a quantitative lack-of-fit measure. It is strongly regime dependent. For in-degrees of the communication networks the discrepancy is large (`email-Eu`: 2.01 vs. 4.21; `CollegeMsg`: 2.15 vs. 4.61), because $\hat{\beta} \to 1$ drives c_1 to a value that places the MLE-implied exponent near 2 whatever the data tail looks like. For growing networks it falls by about an order of magnitude without disappearing: in-degree `Reddit` 2.09 vs. 2.37 and `MathOverflow` 2.41 vs. 2.89; out-degree 2.13 vs. 2.63 and 2.14 vs. 2.97. The large gap for the communication networks is thus a sign that the model is applied outside its natural domain rather than an estimation problem, while some misfit persists for growing networks. In the two smallest networks $M^{1/2}$ is below the floor of 50, so the Hill values 4.21 and 4.61 use fifty observations and are noisy; the gap still exceeds 2 there, well beyond typical error at this sample size.

7.2 Out-of-sample predictive validation

Setup. The history is cut in time: the first 70% of edges (up to n_1) are used to compute $\hat{\theta}$. From the frozen graph $G(n_1)$ we then run $R = 40$ simulations of the model with parameters $\hat{\theta}$ up to the final edge count n_2; this is legitimate because [1] allows any finite starting graph. The tail cdf of the real $G(n_2)$ is plotted against the pointwise 5th–95th percentile envelope of the simulations, and the gap is summarised by the mean KS distance.

A low KS value is not meaningful by itself, as 70% of the final graph coincides with the data by construction. We therefore add a *baseline*, $\mathrm{KS}(G(n_1), G(n_2))$, the error of the forecast "nothing changes", which ignores the model entirely. The model contributes only to the extent that its KS is below that of the baseline.

Finding. Taken at face value, Figure 4 suggests a clear ranking: for `email-Eu` the real cdf leaves the envelope in the middle of the distribution (in-degree KS = 0.235), `CollegeMsg` is closer (0.102), and for the expanding `Reddit` and `MathOverflow` the data follow the envelope over almost four decades, with KS as low as 0.019.

The baseline alters this reading. Its in-degree KS values are 0.081, 0.036, 0.003, 0.025 in the same order, so the ranking mostly measures how much each network moves during the test window. In three networks the model is *worse* than "nothing changes" (`Reddit` 0.019 vs. 0.003, `email-Eu` 0.235 vs. 0.081, `CollegeMsg` 0.102 vs. 0.036). The only clear gain is for the in-degree of `MathOverflow`, 0.019 vs. 0.025.

Varying the training fraction (Table 4) does not change the conclusion. Both distances increase as the forecast horizon lengthens, but their ordering is stable: the baseline is better for `email-Eu`, `CollegeMsg` and `Reddit` at all splits, the model is better for `MathOverflow` in-degree at 0.5 and 0.7, and at 0.9 the test window is too short and the two tie at 0.007.

The cause is the statistic, not a defect of the model. In these networks the degree-tail cdf

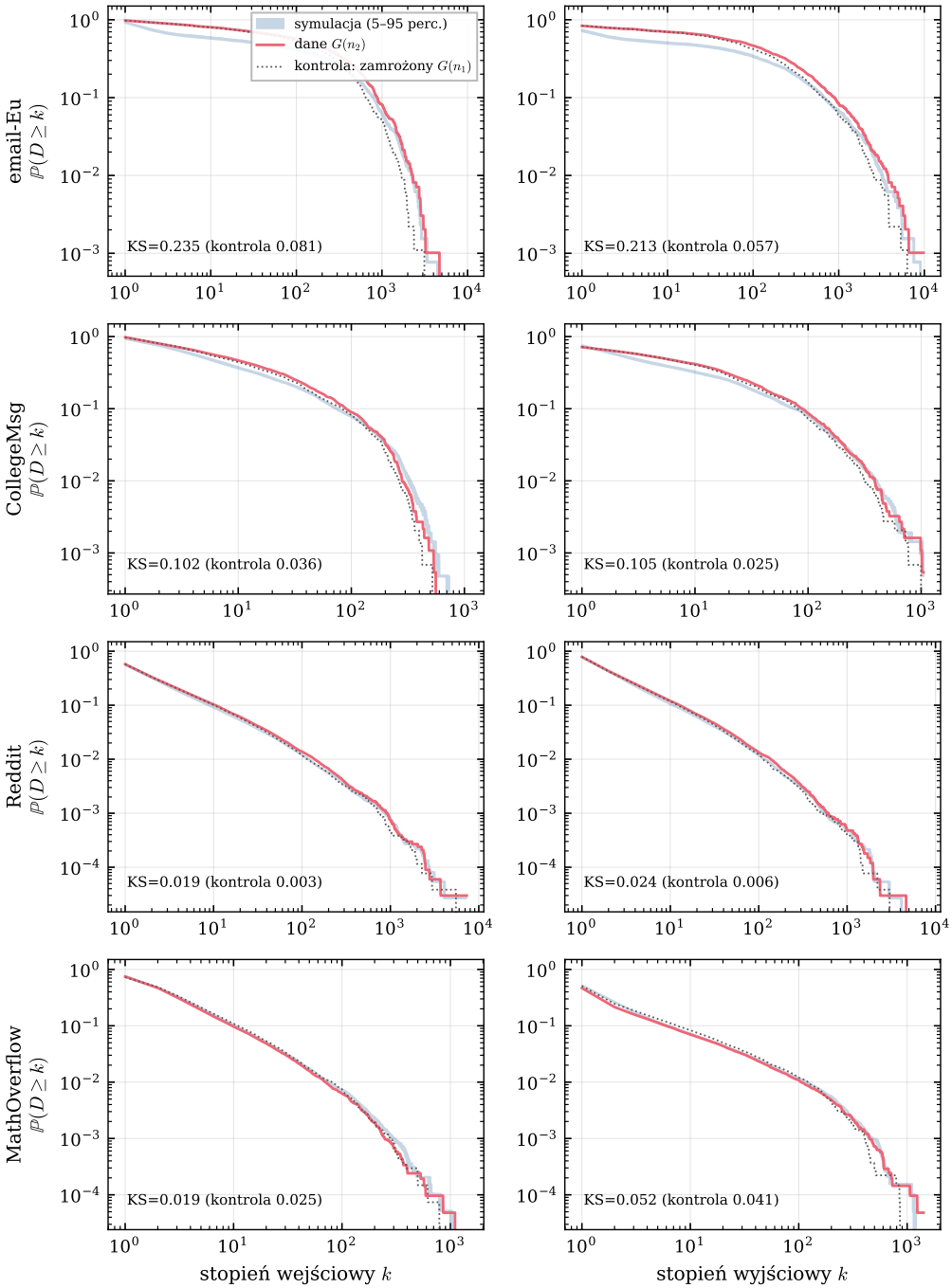


Figure 4: Forecast check for the four networks (rows, from closed to growing) for in-degree (left) and out-degree (right). Solid: tail cdf of the real $G(n_2)$. Band: 5–95 percentile envelope of $R = 40$ simulations from $G(n_1)$ using $\hat{\theta}$ from the first 70% of the history. Dotted: baseline, the frozen $G(n_1)$. Each panel lists the model KS and, in brackets, the baseline KS. Log–log axes.

Table 4: KS distance in the forecast check for three training fractions m/n. Each entry is the model value followed, after a slash, by the baseline ("nothing changes"). Bold marks cases where the model beats the baseline.

	$\mathrm{KS}_{\mathrm{in}}$ (model / baseline)			$\mathrm{KS}_{\mathrm{out}}$ (model / baseline)		
network	0.5	0.7	0.9	0.5	0.7	0.9
`email-Eu`	0.378 / 0.150	0.235 / 0.081	0.081 / 0.036	0.343 / 0.106	0.212 / 0.057	0.076 / 0.030
`CollegeMsg`	0.174 / 0.066	0.101 / 0.036	0.025 / 0.017	0.180 / 0.049	0.104 / 0.025	0.036 / 0.017
`Reddit`	0.029 / 0.008	0.019 / 0.003	0.008 / 0.003	0.042 / 0.012	0.024 / 0.006	0.008 / 0.003
`MathOverflow`	**0.022** / 0.046	**0.019** / 0.025	0.007 / 0.007	0.105 / 0.090	0.052 / 0.041	0.015 / 0.010

barely moves in time: $G(n_1)$ and $G(n_2)$ have almost the same degree distribution although the network keeps growing (`Reddit`: 26,031 → 33,488 vertices). On this statistic the frozen graph is a very strong forecast, whereas model simulations reproduce the right shape but add random variation of their own. KS on degree distributions is thus a poor discriminator. The baseline also is not a model of the growing network: it fixes $N(n_2) = N(n_1)$ and therefore misses the vertex count by tens of percent, something a shape-based statistic cannot register.

7.3 Stability of parameters over time

The forecast check presumes that parameters are constant along the trajectory. To test this, $\hat{\theta}$ is re-estimated on growing prefixes of each history, with the horizontal axis given as the fraction m/n so that networks of different sizes can be compared.

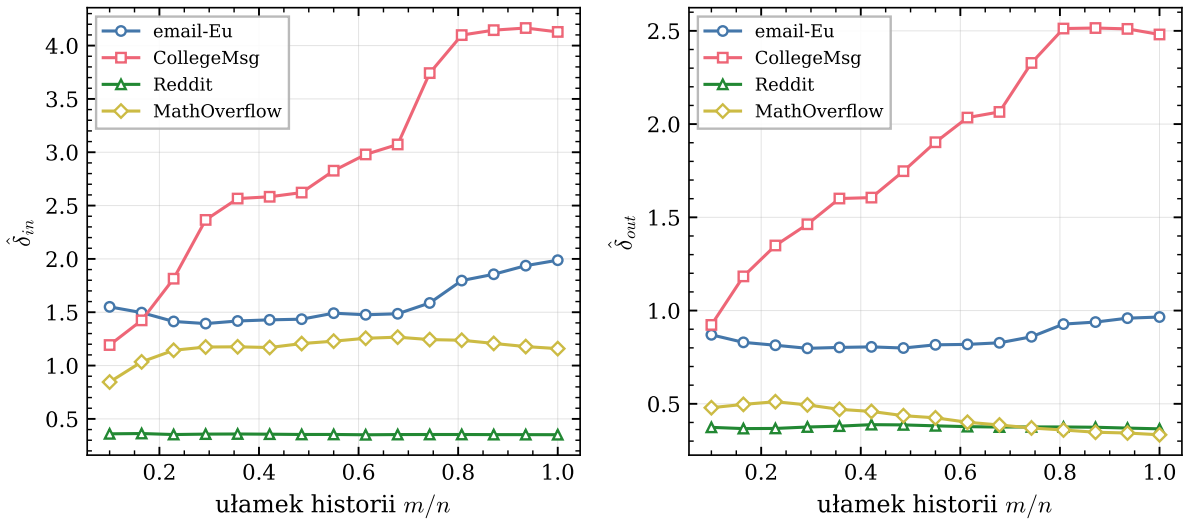


Figure 5: Estimates $\hat{\delta}_{\mathrm{in}}$ (left) and $\hat{\delta}_{\mathrm{out}}$ (right) computed on growing prefixes of the history, for the four networks, against the common axis m/n.

The networks differ markedly in Figure 5. `Reddit` is almost constant ($\hat{\delta}_{\mathrm{in}}$: 0.360 → 0.352; $\hat{\delta}_{\mathrm{out}}$: 0.374 → 0.366), `MathOverflow` drifts moderately and smoothly ($\hat{\delta}_{\mathrm{in}}$: 0.85 → 1.16; $\hat{\delta}_{\mathrm{out}}$: 0.48 → 0.33), and `email-Eu` lies in between, with a modest increase near the end ($\hat{\delta}_{\mathrm{in}}$: 1.55 → 1.99). `CollegeMsg` shows a steady rise ($\hat{\delta}_{\mathrm{in}}$: about 1.2 → 4.1; $\hat{\delta}_{\mathrm{out}}$: 0.9 → 2.5), i.e. non-stationarity: the attachment mechanism evolves, and early-history estimates do not describe later phases. [2] report comparable behaviour for a Wikipedia network.

Stable shifts do not mean a stable θ. Even in `Reddit`, $\hat{\beta}$ grows from 0.76 to 0.88, meaning that new vertices arrive ever less often.

8 Conclusions

Synthetic networks. The simulator reproduces the theory of [1]: exact limiting constants p_i, q_j and tail exponents. Downward bias of the Hill estimates is attributable to the limited width of the power-law window and does not contradict the theorem. Fluctuations of $X_i(t)/t$ are of order $t^{1/2}$, with a possible logarithmic factor that cannot be resolved numerically. The MLE of [2] is efficient, its blocks are uncorrelated as the Fisher matrix predicts, and its intervals are well calibrated; $\hat{\delta}$ becomes approximately normal at about $3 \cdot 10^4$ edges, roughly ten times later than $\hat{\alpha}, \hat{\beta}$, although coverage is already adequate at 10^3 edges.

Real networks. Exponent discrepancy, absolute KS distance and drift of $\hat{\delta}$ all order the four networks in the same way, by the share of schemes A and C. The closed communication networks ($\hat{\beta} \approx 1$) fit poorly. In the expanding `Reddit` and `MathOverflow` networks the exponent discrepancy is an order of magnitude smaller, the shifts are stable, and simulated degree distributions match the data to about 0.02 in KS. The baseline comparison tempers this: the model improves on "nothing changes" in only one of eight cases. The model can therefore be called adequate, in the sense of exponent consistency and stable shifts, chiefly for networks with substantial vertex growth, whereas its predictive edge over a trivial forecast of the degree distribution is slight. Part of this stems from the metric, part from non-stationary parameters that appear even in the best-fitting networks, a caveat also raised in [2].

Limitations and extensions. Natural continuations are the full model with schemes creating two new vertices, time-varying parameters for non-stationary networks, and validation with statistics that react more strongly to growth than the marginal degree distribution, such as the joint in/out-degree distribution.

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